# WAYA ROBOTIC COMPETITION 2.0

# OFFICIAL RULE BOOK

## UNIVERSITY CATEGORY





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WAYA BATTLE BOTS

## 1. Competition Purpose

WAYA Robotics Competition 2.0 is a national-level robotics competition designed to promote robotics, innovation, practical engineering skills, creative problem solving, and hands-on technical exposure among students in Sri Lanka.

## 2. Eligibility

- The University Category is open to recognized universities in Sri Lanka.
- Each participant must be a registered student of the school or university represented by the team.

## 3.Team Composition

- Minimum team size: 1 member.
- Maximum team size: 5 members.
- All team members must belong to the institution represented by the team.

## 4.Robot Eligibility and General Technical Rules

- Only fully autonomous robots are permitted.
- Maximum robot size (length x width) :25 cm × 20 cm.
- Internal power supply must not exceed 24 V.
- Remote control of the robot is not allowed.
- The robot must include a kill switch.
- The robot must not damage the arena or competition equipment.
- Remote AI agents must not be used on development boards.
- Each team will be given a maximum of **one minute (1 minute)** for robot calibration before the official run begins.
- No major hardware modifications, code uploading, or replacement of robot components are allowed during the calibration period.
- Robot must be handover to the organizing committee at the registration desk.
- Detailed technical guidelines will be shared with participants by the organizers.

## 5.General Arena Information

- Arena background: white.
- Arena line color: black.
- Line width: 3 cm.
- Object size 5cm * 5cm * 5cm
- Arena sections are tentatively 12 ft × 6 ft; dimensions may change with future developments.
- Safe and durable materials will be used.
- The organizers may make necessary changes to the arena before the competition.

## 6.Safety

- Teams must operate robots responsibly and follow organizer instructions.
- The robot must have a functional kill switch.
- Robots must not intentionally or negligently damage the arena.
- Electrical and technical safety instructions issued by the organizers must be followed.

## 7.Judging and Final Authority

- Total score: 100 marks.
- Task Completion: 40 marks.
- Speed Performance: 30 marks.
- Completion Consistency: 15 marks.
- Judges’ Evaluation: 15 marks.
- The judges’ decision is final.

## 8. Category-Specific Challenge

The University is designed as a advanced robotics challenge. The following tasks are identified in the official competition proposal:

- Line following.
- Sharp turns, including 45,90-degree turns.
- Obstacle avoidance.
- Wall navigation.
- Uncompleted line detection.
- **Color detection and object pick and drop task**
    - **1. Description**
        - **The robot is given Red or Blue before the run.**
        - **At the Color Selection Zone, the track splits into a Red path (left) and a Blue path (Right). The robot follows only the assigned color path to the Pick Point.**
        - **It picks up the box, carries it to the Drop Point, releases it, then goes to the End Point.**
    - **2. Color Assignment**
        - **Judges choose the color by random draw before each run.**
        - **The team enters it once, before the start (button or switch). No remote or wireless selection.**
    - **3. Specifications**
        - **Box: 5 × 5 × 5 cm, max 70 g. One box at the end of each colored path.**
        - **Drop area: 20 × 20 cm marked square.**
        - **End Point: marked area where the robot must stop.**
    - **4. Rules**
        - **The task must be done autonomously.**
        - **Entering the wrong color path: −5 points.**
        - **The box must be carried, not pushed or dragged.**
        - **The box must be fully inside the drop area when released.**
        - **The robot must reach the End Point to complete the run.**
- **Ramp Obstacle**
    - **1. Description**
        - The track includes one ramp. The robot must **climb and descend** it while following the line.
    - **2. Specifications**
        - Angle: **15°** (±2°)
        - Length: **60 cm**
        - Width: **30 cm** (same as track)
        - Top platform: **30 × 30 cm**
        - Surface and line: same as the main track
        - No side walls.
    - **3. Rules**
        - The line is **continuous** across the ramp.
        - The robot must cross **autonomously**, with no manual help.
        - If the robot falls off, restart from the **last checkpoint** with a **+10 s penalty**.
        - Grip and wheel slip are the **team's responsibility**.

### Task and Arena Notice

The final tasks, exact arena layout, and detailed technical guidelines will be officially released by the organizers before the competition. The proposal states that arena arrangements may be changed if necessary.

### Category Notes

- The proposal describes this as an advanced robotics challenge focused on technical complexity and innovation.
- Final tasks and arena layout will be officially released prior to the competition.
- The University Category arena may be changed if necessary.

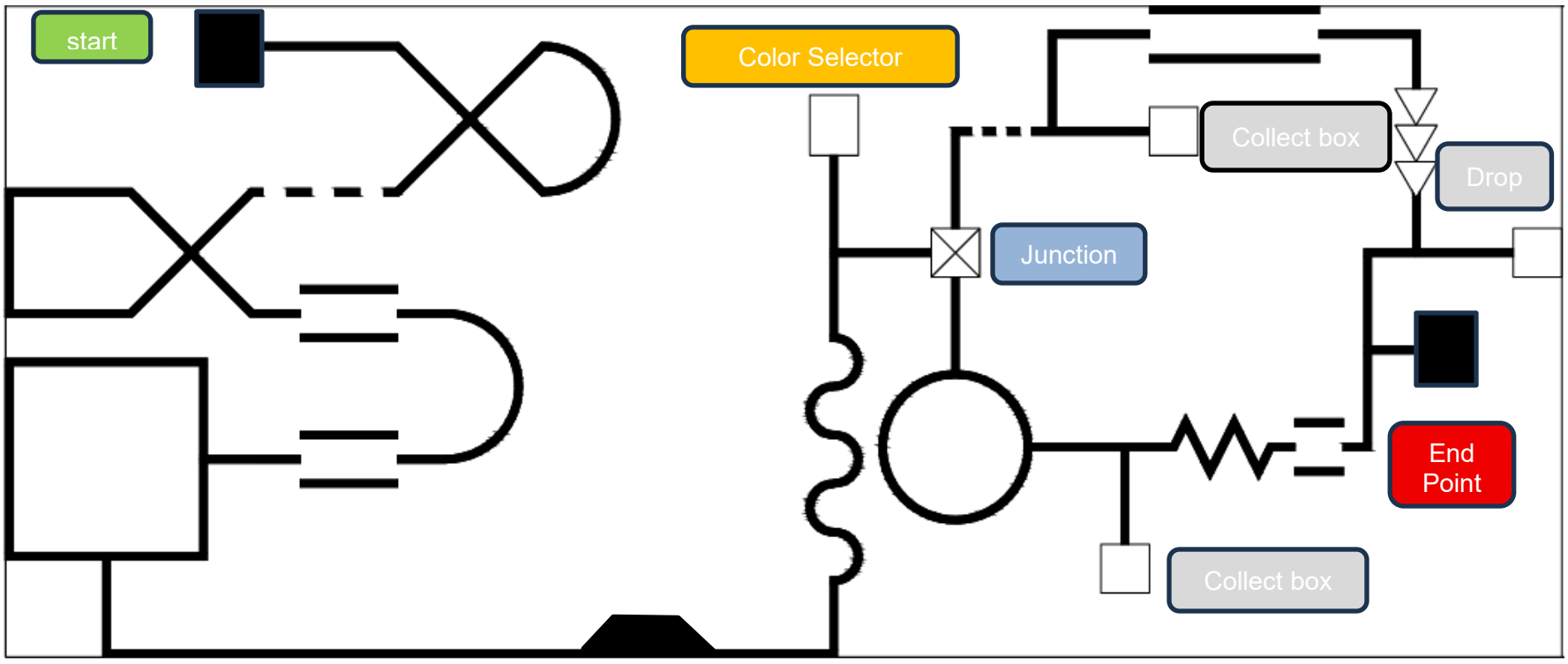
## 9.Competition Procedure

- Teams must submit their robots during the official robot submission period.
- Each team will be given **three (3) official runs** during the competition.
- The best-performing run out of the three attempts will be considered as the team's **final result**.
- Robots may be inspected for compliance with the published technical requirements.
- The best run will be determined based on the published scoring categories (Task Completion, Speed Performance, and Completion Consistency).
- Teams must follow instructions given by technical officials and judges.
- Only eligible autonomous robots may participate in official scoring runs.
- Competition runs will be conducted according to the finalized arena and schedule.
- The organizers may stop a run for safety, technical, or arena-related reasons.
- The judging panel will record scores according to the published scoring categories.

### Published Event Schedule

| Activity | Time |
|---|---|
| Robot submission | 8:30 am – 10:00 am |
| Welcome speech | 10:00 am – 10:15 am |
| Competition starts | 10:15 am |
| Refreshments | 11:30 am |
| Competition continues | 11:45 am |
| Lunch | 1:30 pm – 2:10 pm |
| Competition ends | 4:30 pm |
| Prize awarding / certificates | 5:15 pm |
| Event ends | 6:00 pm |

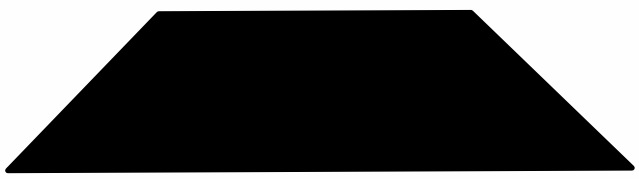
## 10. Arena Map and Obstacle Layout (Subject to change)




### Ramp Obstacle

Specifications

- Angle: 15° (±2°)
- Length: 60 cm
- Width: 30 cm (same as track)
- Top platform: 30 × 30 cm
- Surface and line: same as the main track
- No side walls.



### Wall Navigation

Specifications

- Section length: 40 cm
- Wall height: 10 cm
- Gap between walls: 30 cm
- Floor: white, no line
- Walls: black, solid color



## 11. Awards and Recognition

| Place | Recognition | Cash Prize |
|---|---|---|
| 1st | Championship Trophy + Official Certificate | Rs. 70,000 |
| 2nd | Trophy + Official Certificate | Rs. 50,000 |
| 3rd | Trophy + Official Certificate | Rs. 30,000 |

•Participation e-certificates will be awarded to all registered teams that successfully compete.

## 12. Organizer Authority and Changes

• The organizers may update the arena, task details, or technical instructions when necessary.

• Any official changes communicated to registered teams become part of the competition requirements.

• Judges' decisions on competition scoring are final.

• Participants are responsible for checking official announcements before the event.

## 13. Administration

The competition is organized by the Waya Robotics Club (WRC) and Wayamba Engineering Technology Society (WETS), Wayamba University of Sri Lanka, with academic guidance from the Faculty of Technology.

Main Advisor: Dr. I.P.M. Wickramasinghe, Dean, Faculty of Technology, Wayamba University of Sri Lanka.